

## EfficientNet-B2 Fire / No-Fire Report

Model: EfficientNet-B2  
Image Size: 260x260  
Batch Size: 32  
Optimizer: AdamW  
Loss: Focal Loss (gamma=2.0)  
Checkpoint Path: /kaggle/working/b2\_fire\_best\_model.pth

Dataset Split:  
Train size: 4620  
Validation size: 990  
Test size: 991  
Total size: 6601

Class Mapping:  
fire = 0  
no\_fire = 1

Loaded Checkpoint:  
Checkpoint epoch: 15  
Best Validation Fire F1: 1.0000

TEST RESULTS:  
Test Loss: 0.0078  
Test Accuracy: 0.9980  
Test Macro F1: 0.9922

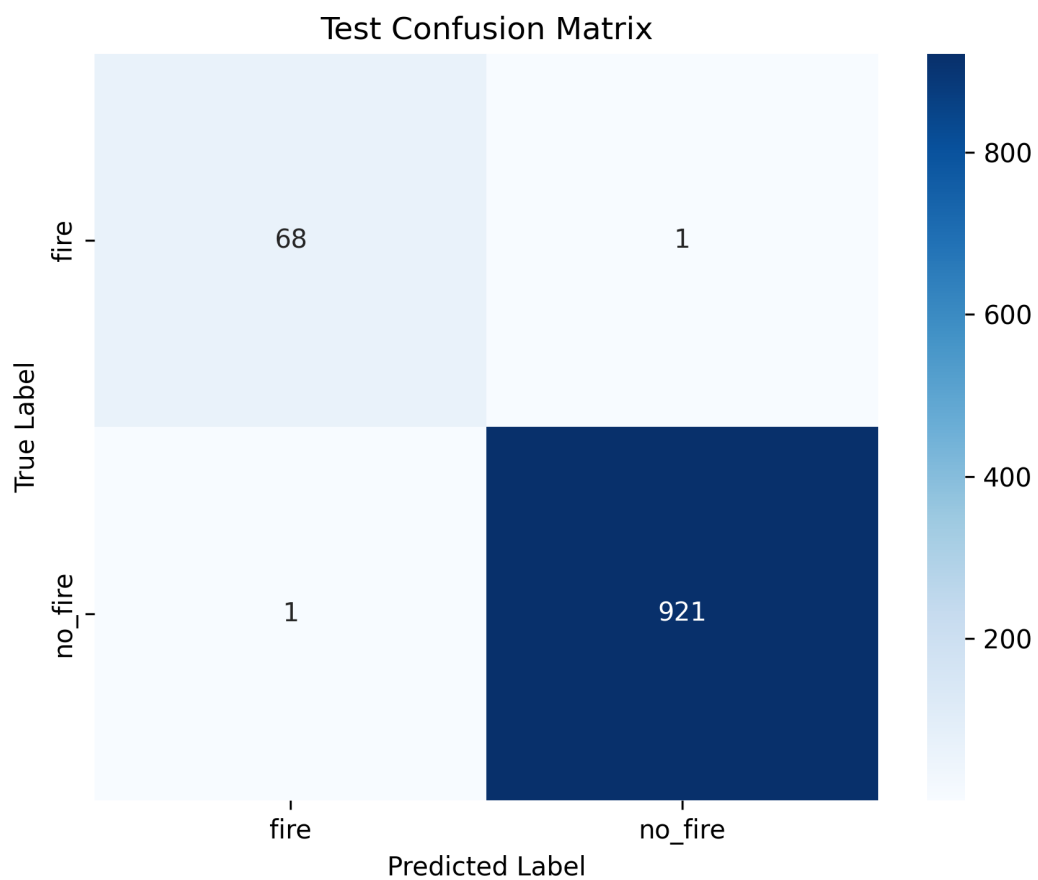
Fire Class:  
Precision: 0.9855  
Recall: 0.9855  
F1-score: 0.9855

No-Fire Class:  
Precision: 0.9989  
Recall: 0.9989  
F1-score: 0.9989

## Classification Report - Test Set

| class        | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| fire         | 0.9855    | 0.9855 | 0.9855   | 69      |
| no_fire      | 0.9989    | 0.9989 | 0.9989   | 922     |
| accuracy     |           |        | 0.9980   | 991     |
| macro avg    | 0.9922    | 0.9922 | 0.9922   | 991     |
| weighted avg | 0.9980    | 0.9980 | 0.9980   | 991     |

# Test Confusion Matrix



## Model Training Report

Model: EfficientNet-B2

Image Size: 260x260

Batch Size: 32

Optimizer: AdamW

Loss: Focal Loss

Primary Metric: Validation Fire F1-score

Best Validation Result:

Stage: stage2

Epoch: 15

Val Fire F1: 1.0000

Val Macro F1: 1.0000

Val Accuracy: 1.0000

Val Loss: 0.0002

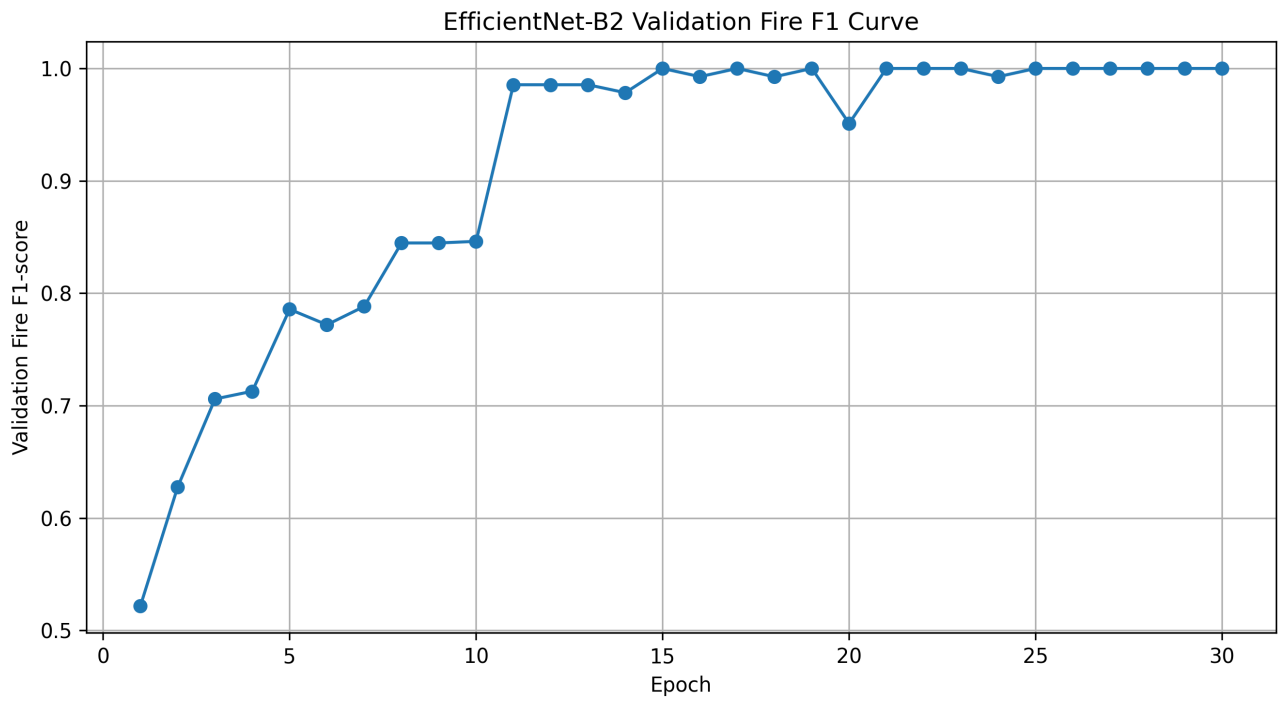
Best model path:

/kaggle/working/b2\_fire\_best\_model.pth

# Epoch Results

| stage  | epoch | train_loss | val_loss | val_accuracy | val_macro_f1 | val_fire_f1 | time_sec |
|--------|-------|------------|----------|--------------|--------------|-------------|----------|
| stage1 | 1     | 0.4320     | 0.2488   | 0.9000       | 0.7330       | 0.5217      | 79.0533  |
| stage1 | 2     | 0.1710     | 0.1714   | 0.9293       | 0.7943       | 0.6277      | 68.6670  |
| stage1 | 3     | 0.1089     | 0.1367   | 0.9444       | 0.8376       | 0.7059      | 68.8549  |
| stage1 | 4     | 0.0755     | 0.1308   | 0.9455       | 0.8413       | 0.7128      | 64.9654  |
| stage1 | 5     | 0.0636     | 0.0765   | 0.9636       | 0.8829       | 0.7857      | 67.6592  |
| stage1 | 6     | 0.0432     | 0.0688   | 0.9606       | 0.8752       | 0.7719      | 65.9353  |
| stage1 | 7     | 0.0438     | 0.0710   | 0.9636       | 0.8842       | 0.7882      | 62.9973  |
| stage1 | 8     | 0.0399     | 0.0528   | 0.9747       | 0.9155       | 0.8447      | 73.3217  |
| stage1 | 9     | 0.0328     | 0.0540   | 0.9747       | 0.9155       | 0.8447      | 66.4477  |
| stage1 | 10    | 0.0247     | 0.0388   | 0.9758       | 0.9165       | 0.8462      | 68.6859  |
| stage2 | 11    | 0.0236     | 0.0023   | 0.9980       | 0.9922       | 0.9855      | 76.6759  |
| stage2 | 12    | 0.0188     | 0.0020   | 0.9980       | 0.9922       | 0.9855      | 75.6885  |
| stage2 | 13    | 0.0020     | 0.0030   | 0.9980       | 0.9922       | 0.9855      | 76.8152  |
| stage2 | 14    | 0.0016     | 0.0070   | 0.9970       | 0.9884       | 0.9784      | 75.1988  |
| stage2 | 15    | 0.0011     | 0.0002   | 1.0000       | 1.0000       | 1.0000      | 80.9591  |
| stage2 | 16    | 0.0019     | 0.0002   | 0.9990       | 0.9961       | 0.9927      | 73.9368  |
| stage2 | 17    | 0.0015     | 0.0000   | 1.0000       | 1.0000       | 1.0000      | 76.4385  |
| stage2 | 18    | 0.0044     | 0.0005   | 0.9990       | 0.9960       | 0.9926      | 76.6112  |
| stage2 | 19    | 0.0054     | 0.0001   | 1.0000       | 1.0000       | 1.0000      | 73.0287  |
| stage2 | 20    | 0.0073     | 0.0187   | 0.9929       | 0.9736       | 0.9510      | 78.9435  |
| stage2 | 21    | 0.0034     | 0.0000   | 1.0000       | 1.0000       | 1.0000      | 78.6593  |
| stage2 | 22    | 0.0014     | 0.0001   | 1.0000       | 1.0000       | 1.0000      | 76.5784  |
| stage2 | 23    | 0.0047     | 0.0000   | 1.0000       | 1.0000       | 1.0000      | 75.9468  |
| stage2 | 24    | 0.0019     | 0.0018   | 0.9990       | 0.9961       | 0.9927      | 78.6480  |
| stage2 | 25    | 0.0000     | 0.0001   | 1.0000       | 1.0000       | 1.0000      | 81.1715  |
| stage2 | 26    | 0.0000     | 0.0001   | 1.0000       | 1.0000       | 1.0000      | 80.2102  |
| stage2 | 27    | 0.0000     | 0.0000   | 1.0000       | 1.0000       | 1.0000      | 77.3508  |
| stage2 | 28    | 0.0009     | 0.0000   | 1.0000       | 1.0000       | 1.0000      | 77.8997  |
| stage2 | 29    | 0.0070     | 0.0000   | 1.0000       | 1.0000       | 1.0000      | 75.2426  |
| stage2 | 30    | 0.0060     | 0.0000   | 1.0000       | 1.0000       | 1.0000      | 79.7332  |

# Validation Fire F1 Over Epochs



# Train Loss vs Validation Loss

